Exercise - 13

Implement your own ping program

Aim:

To implement a ping program python to check host reachability and calculate round-trip time.

Algorithm:

1. Start the program
2. Get the hostname or IP address
3. Create a raw ICMP socket
4. Construct an ICMP echo request packet and calculate its checksum.
5. Send the packet and wait for a reply.
6. Calculate and display RTT
7. Repeat four times and terminate

Program:

```
import socket
import struct
import time
def checksum (data):
    total = 0
    for i in range (0, len (data) - 1, 2):
        total += (data[i] << 8) + data[i+1]

    if len (data) % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0xffff)
    total += total >> 16
    return (~total) & 0xffff
```

sample output:

Enter hostname or IP address:
8.8.8.8
Reply from 8.8.8.8 : bytes = 43
time = 24.51 ms
Reply from 8.8.8.8 bytes = 43
time = 24.12 ms
Reply from 8.8.8.8 bytes = 43
time = 24.36 ms

```
socket.getprotobyname("icmp"))
sock.settimeout(2)
pid = os.getpid() & 0xffff
data = b"computer Networks Ping"
header = struct.pack("!BBHHH", 8, 0, 0, pid, 1)
start = time.time()
sock.sendto(packet, (addr, 1))
rtt = (time.time() - start) * 1000
except socket.timeout:
    print("Request timed out")
except PermissionError:
    print("Run the administrator / root privileges")
host = input("Enter hostname or IP address: ")
for i in range(4):
    ping(host)
    time.sleep(1)
```

Result:

Thus, the ping program was implemented successfully in python to check network connectivity and calculate RTT.